

# SAFIR: A Sub-Dialectal Benchmark for Saudi Arabic–English Machine Translation

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## Abstract

Arabic machine translation benchmarks provide limited coverage of variation within Saudi Arabic. We introduce SAFIR, a Saudi-centric benchmark of 610 unique sentence pairs organized into 12 creator-specific evaluation splits across five regional groups: Najdi, Hijazi, Southern, Northern, and Eastern. Native speakers authored the Arabic sentences and their English translations; 235 pairs received independent double review for dialect authenticity, translation adequacy, and fluency. We evaluate twelve systems, comprising closed-weight LLMs, open-weight LLMs, and dedicated MT models, in the zero-shot Saudi→English setting using reference-based metrics and GPT-4o judgments. Closed-weight LLMs obtain the strongest aggregate results. Gemini 2.5 Pro achieves the highest average BLEU (34.41), while GPT-5 achieves the highest BERTScore (83.60). Rankings vary across splits, and Southern varieties present particular challenges in judged adequacy. Differences between reference overlap, adequacy, and fluency support reporting performance by sampled variety and evaluation dimension. SAFIR and its leaderboard support more detailed assessment of Saudi Arabic translation.

## 1 Introduction

Machine translation has advanced substantially for Modern Standard Arabic (MSA), but evaluating the spoken varieties used in everyday communication remains challenging. Dialectal resources represent only part of the lexical, morphological, and phonological variation found across Arabic-speaking communities. Evaluation under broad country or regional labels can further conceal differences between local varieties.

Saudi Arabic illustrates the need for finer coverage. Linguistic descriptions distinguish regional groupings such as Central (Najdi), Western (Hijazi), Southern, Eastern, and Northern, with further variation across cities and communities (Al-Monef et al.,

2026). Existing resources capture parts of this diversity. MADAR, for example, includes Riyadh and Jeddah and already supports city-level Saudi comparisons (Bouamor et al., 2018). Coverage across a broader range of Saudi locations nevertheless remains limited.

We introduce SAFIR, a Saudi-centric machine translation benchmark comprising 610 unique Saudi–English sentence pairs. Its collection design spans eight locations in seven administrative regions, organized into five regional groups and 12 creator-specific evaluation splits. Native speakers authored and translated the sentences, and an independent quality assessment double-reviewed 235 pairs. The splits preserve the individual collections and their local labels; they do not constitute a validated taxonomy of twelve distinct dialects. This distinction enables detailed evaluation while keeping the linguistic interpretation of split-level differences appropriately scoped.

Using SAFIR, we evaluate twelve systems comprising closed-weight LLMs, open-weight LLMs, and dedicated MT models in the zero-shot Saudi→English setting. We combine reference-based metrics with LLM judgments to examine complementary aspects of translation quality. The public leaderboard provides aggregate and split-level comparisons.<sup>1</sup> Our contributions are:

- A Saudi-centric benchmark with native-speaker authorship and translation, coverage across five regional groups, and independent double review of a 235-pair quality-control sample.
- Evaluation of twelve systems across creator-specific collections, using complementary measures of reference overlap and judged quality.

<sup>1</sup><https://huggingface.co/spaces/NAMAA-Space/SAFIR-Leaderboard>

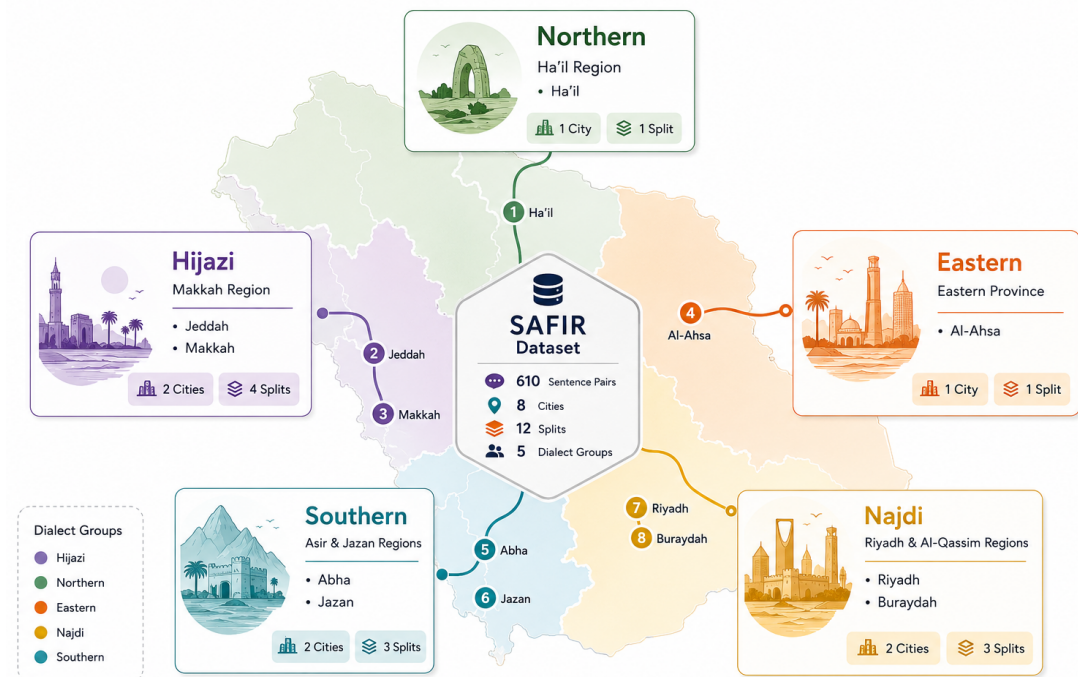


Figure 1: SAFIR’s collection design across five regional groups and eight designated collection locations in seven administrative regions. The 12 creator-specific evaluation splits contain 610 unique Saudi–English pairs. Locations are schematic and not to scale; colors indicate regional groups.

- A public leaderboard for comparing Saudi Arabic translation performance at aggregate and split levels.

## 2 Related Work

Arabic MT resources differ in their geographic coverage and construction. PADIC aligns Algerian, Tunisian, Levantine, and MSA varieties, without a Saudi or Gulf variety (Meftouh et al., 2015). WMT24++ extends evaluation to 55 languages and dialects, but represents Saudi Arabic through a single ar\_SA track (Deutsch et al., 2025).

Other resources provide regional or city labels. FLORES+ includes approximately 2,009 Najdi sentences across its development and devtest partitions (NLLB Team, 2022). MADAR covers 25 Arabic cities, including Riyadh and Jeddah with 2,000 sentences per Saudi city (Bouamor et al., 2018). Alexandria provides human translations and peer revision across dialects from 13 Arab countries. Its Saudi material uses Southern, Hijazi, and Khaleeji labels and also includes city-of-origin metadata (El Mekki et al., 2026).

These resources already support country, regional, or city-level evaluation. SAFIR complements them through focused coverage across Saudi regional groups and separate evaluation of local cre-

ator collections. Its contribution is the breadth of its Saudi collection design and its evaluation granularity, rather than the first use of city labels for Saudi Arabic. Table 1 compares Saudi coverage, subset sizes, reference construction, and human involvement across these resources.

## 3 The SAFIR Benchmark

Construction proceeded through Arabic sentence authorship, English translation, and human quality review over approximately six months of community work. The benchmark contains 610 unique Saudi–English sentence pairs.

### 3.1 Dialectal and Regional Coverage

The collection design covers seven of Saudi Arabia’s 13 administrative regions:<sup>2</sup> Riyadh, Al-Qassim, Ha’il, Makkah, Asir, Jazan, and the Eastern Province. Its eight designated collection locations are Riyadh, Buraydah, Ha’il, Makkah, Jeddah, Abha, Jazan, and Al-Ahsa. We organize the material into Najdi, Hijazi, Northern, Southern, and Eastern groups; Southern combines the sampled Asir and Jazan material.

Each evaluation split contains the material of

<sup>2</sup>ISO 3166-2:SA: <https://www.iso.org/obp/ui/#iso:code:3166:SA>.

| Dataset                               | Saudi coverage                                                                | Saudi subset size            | Creation direction         | Human translation and review                                                        |
|---------------------------------------|-------------------------------------------------------------------------------|------------------------------|----------------------------|-------------------------------------------------------------------------------------|
| WMT24++<br>(Deutsch et al., 2025)     | One country-level label: ar_SA                                                | 998 segments                 | En→Saudi                   | Professional translation and post-editing; sampled dialect checks.                  |
| PADIC<br>(Meftouh et al., 2015)       | No Saudi variety                                                              | 0                            | Ar↔Ar, with MSA as a pivot | Native-speaker translation; separate review not specified.                          |
| FLORES+<br>(NLLB Team, 2022)          | One regional variety: Najdi                                                   | 2,009 sentences              | MSA→Najdi; aligned with En | Human adaptation under the FLORES quality-assurance protocol.                       |
| MADAR<br>(Bouamor et al., 2018)       | Two Saudi cities: Riyadh and Jeddah                                           | 4,000 pairs (2,000 per city) | En/Fr→Saudi                | Native-speaker translation; separate corpus review not specified.                   |
| Alexandria<br>(El Mekki et al., 2026) | Three dialect labels: Southern, Hijazi, and Khaleeji; city-of-origin metadata | 11,807 turns                 | En→Saudi                   | Human translation and peer revision.                                                |
| <b>SAFIR</b>                          | <b>Five regional groups with 12 creator-specific evaluation splits</b>        | 610 unique pairs             | Saudi→En                   | Native-speaker authorship and translation; 235 pairs independently double-reviewed. |

Table 1: Expanded comparison of Saudi coverage and construction. En, Fr, and Ar denote English, French, and Arabic. Creation direction concerns reference construction, not every possible evaluation direction. WMT24++ collected 998 segments and retained 960 for evaluation. FLORES+ counts dev and devtest; MADAR’s two Saudi varieties share 2,000 source sentences. Alexandria’s size is its Saudi total across partitions. SAFIR counts unique Arabic–English pairs.

one creator (Table 2). Same-city labels distinguish collections: Najdi Riyadh and Najdi Urban were assigned to Riyadh; Hijazi Makkah and Hijazi Urban to Makkah; and Hijazi Jeddah and Hijazi Urban Jeddah to Jeddah. “Urban” was retained from collection and is not an operationally defined linguistic category. Likewise, Southern Qahtan and Southern Asiri retain community and regional descriptors without establishing mutually exclusive linguistic categories. No predetermined set of linguistic features was used to separate these collections. Split comparisons therefore reflect sampled language use, creator choices, and reference style alongside possible dialectal differences.

### 3.2 Arabic Sentence Collection

Sixteen native-speaker participants contributed: eight were creators only, four reviewers only, and four served in both roles. Thus, there were twelve creators and eight distinct reviewers. No participant reviewed their own material. The creators represented Najdi (three), Northern (one), Hijazi (four), Southern (three), and Eastern (one). Eligibility required residence in the target location, fluency in the local variety, and proficiency in English.

Shared guidelines asked each creator to author approximately 50 conversational sentences containing vocabulary characteristic of the assigned variety. This was a flexible target. Creators composed sen-

tences from their knowledge of everyday language in their communities and checked their local authenticity. Copying from published datasets was prohibited. This authorship requirement reduces deliberate reuse but does not establish the absence of incidental overlap with prior corpora or model training data.

### 3.3 English Translation

Each creator translated their own sentences into English, aiming to preserve the source meaning and communicative intent. Machine translation tools were permitted as an aid, but creators remained responsible for the final references.

### 3.4 Human Quality Review

We sampled up to 20 sentences per creator, yielding 235 pairs for independent double review. Two reviewers were assigned to each regional group, with reviewer assignments shared across groups where needed. The sampling totals were Najdi (59), Northern (20), Eastern (20), Southern (57), and Hijazi (79). Reviewers used a 1–5 ordinal scale for *dialect authenticity* (reflection of the claimed variety), *translation adequacy* (preservation of meaning), and *translation fluency* (naturalness and grammaticality of the English).

Mean ratings were 4.74 for authenticity, 4.57 for adequacy, and 4.70 for fluency (Table 3). Ratings

| Group    | Admin. region    | Collection location | Split label         | Label basis                        |
|----------|------------------|---------------------|---------------------|------------------------------------|
| Eastern  | Eastern Province | Al-Ahsa             | Eastern Urban       | Creator label within Eastern group |
| Hijazi   | Makkah           | Makkah              | Hijazi Makkah       | City descriptor                    |
| Hijazi   | Makkah           | Makkah              | Hijazi Urban        | Creator label within Makkah        |
| Hijazi   | Makkah           | Jeddah              | Hijazi Urban Jeddah | Creator label within Jeddah        |
| Hijazi   | Makkah           | Jeddah              | Hijazi Jeddah       | City descriptor                    |
| Najdi    | Al-Qassim        | Buraydah            | Najdi Qasim         | Regional descriptor                |
| Najdi    | Riyadh           | Riyadh              | Najdi Riyadh        | City descriptor                    |
| Najdi    | Riyadh           | Riyadh              | Najdi Urban         | Creator label within Riyadh        |
| Northern | Ha'il            | Ha'il               | Northern Hail       | Geographic descriptor              |
| Southern | Asir             | Abha                | Southern Qahtan     | Community descriptor               |
| Southern | Jazan            | Jazan               | Southern Jazan      | Geographic descriptor              |
| Southern | Asir             | Abha                | Southern Asiri      | Regional descriptor                |

Table 2: Designated collection locations and naming basis of SAFIR’s 12 evaluation splits. Every row represents one creator’s collection. The final column explains the naming convention rather than a validated linguistic classification. Split labels match those used in the results tables and on the leaderboard.

| Dimension    | Mean | Ratings $\leq 3$ |
|--------------|------|------------------|
| Authenticity | 4.74 | 7.4%             |
| Adequacy     | 4.57 | 10.6%            |
| Fluency      | 4.70 | 7.4%             |

Table 3: Human quality ratings on a scale from 1 to 5. Each dimension contains 470 ratings from independent double review of 235 sentence pairs.

of 3 or below constituted 7.4%, 10.6%, and 7.4% of the respective individual reviewer ratings, rather than those percentages of sentence pairs. The review was diagnostic, not a filtering pass: no pairs were removed on the basis of these scores. Consequently, the benchmark retains items that received low ratings, and quality statistics describe the reviewed sample rather than every pair.

### 3.4.1 Inter-Annotator Agreement

We report Krippendorff’s ordinal  $\alpha$  (Krippendorff, 2004; Artstein and Poesio, 2008) and Gwet’s  $AC_2$  with quadratic weights (Gwet, 2014). Their chance corrections and weights differ, so we interpret both alongside exact agreement and agreement within one scale point. Appendix C provides definitions and descriptive rates.

Overall  $AC_2$  is 0.930 for authenticity, 0.852 for adequacy, and 0.899 for fluency; corresponding  $\alpha$  values are 0.190, 0.085, and 0.072 (Table 4). Within-one-point agreement ranges from 90.0% to 100.0% for Najdi, Northern, Eastern, and Southern, but from 65.8% to 73.4% for Hijazi. Exact agreement is also lower for Hijazi.

Concentrated ratings can yield low chance-

corrected reliability despite high observed agreement (Gwet, 2008). For Northern authenticity, reviewers agree exactly on 19 of 20 items, with  $AC_2 = 0.997$  and  $\alpha = 0.000$ . This illustrates sensitivity to rating distributions, but does not establish uniform reliability. In particular, Hijazi’s lower descriptive agreement indicates substantive disagreement that merits further review.

### 3.5 Dataset Statistics

Table 5 reports collection sizes, mean lengths, and lexical diversity for each evaluation split. Across the 610 unique pairs, mean lengths are 14.5 Arabic and 22.6 English orthographic tokens; medians are 7 and 11. Arabic lengths range from 1 to 177 tokens, while English lengths range from 1 to 232 tokens. Some entries therefore extend beyond short utterances.

Length differs substantially across the collections. Mean Arabic length ranges from 3.8 tokens in Southern Jazan to 56.3 in Hijazi Urban. Even within Makkah, Hijazi Urban averages 56.3 Arabic tokens compared with 12.2 for Hijazi Makkah. In Jeddah, the corresponding means are 28.9 for Hijazi Urban Jeddah and 11.6 for Hijazi Jeddah. These differences reinforce the interpretation of the split names as creator collections: shared geography does not imply comparable source length or discourse structure.

We measure lexical diversity using HD-D with a fixed sample size of 42 tokens (McCarthy and Jarvis, 2010). Arabic split scores range from 92.7 to 96.1 and English scores from 83.3 to 88.3 on the reported 0–100 scale. This statistic estimates

| Group    | N   | Authenticity |          | Adequacy |          | Fluency |          |
|----------|-----|--------------|----------|----------|----------|---------|----------|
|          |     | $AC_2$       | $\alpha$ | $AC_2$   | $\alpha$ | $AC_2$  | $\alpha$ |
| Najdi    | 59  | 0.988        | −0.054   | 0.964    | 0.223    | 0.994   | −0.035   |
| Northern | 20  | 0.997        | 0.000    | 0.903    | 0.063    | 0.923   | 0.289    |
| Eastern  | 20  | 0.974        | −0.083   | 0.941    | 0.098    | 0.970   | −0.182   |
| Southern | 57  | 0.984        | 0.498    | 0.921    | 0.361    | 0.951   | 0.395    |
| Hijazi   | 79  | 0.776        | 0.050    | 0.637    | −0.176   | 0.727   | −0.182   |
| Overall  | 235 | 0.930        | 0.190    | 0.852    | 0.085    | 0.899   | 0.072    |

Table 4: Inter-annotator agreement by regional group and dimension.  $AC_2$  is Gwet’s agreement coefficient with ordinal weights;  $\alpha$  is Krippendorff’s ordinal alpha.

| Evaluation split    | N   | Mean length (SD) |             | $HD-D \times 100$ |         |
|---------------------|-----|------------------|-------------|-------------------|---------|
|                     |     | Arabic           | English     | Arabic            | English |
| Eastern Urban       | 54  | 5.6 (1.4)        | 8.9 (2.6)   | 94.2              | 84.1    |
| Hijazi Makkah       | 49  | 12.2 (5.9)       | 18.4 (9.3)  | 94.0              | 87.0    |
| Hijazi Urban        | 50  | 56.3 (38.0)      | 85.0 (55.4) | 96.1              | 87.2    |
| Hijazi Urban Jeddah | 51  | 28.9 (20.6)      | 42.9 (29.4) | 94.8              | 86.5    |
| Hijazi Jeddah       | 51  | 11.6 (4.8)       | 22.8 (9.7)  | 95.0              | 88.1    |
| Najdi Qasim         | 51  | 19.9 (27.2)      | 32.8 (44.9) | 95.9              | 88.3    |
| Najdi Riyadh        | 51  | 10.0 (7.6)       | 16.3 (12.8) | 95.5              | 87.3    |
| Najdi Urban         | 52  | 4.3 (1.6)        | 6.3 (2.6)   | 92.7              | 85.4    |
| Northern Hail       | 50  | 5.9 (1.4)        | 8.3 (2.4)   | 93.4              | 85.0    |
| Southern Qahtan     | 53  | 7.5 (4.7)        | 11.9 (7.5)  | 94.5              | 86.4    |
| Southern Jazan      | 50  | 3.8 (1.2)        | 6.6 (2.3)   | 94.6              | 83.3    |
| Southern Asiri      | 50  | 8.6 (8.6)        | 12.4 (11.7) | 93.2              | 87.2    |
| Unique pairs        | 610 | 14.5 (20.8)      | 22.6 (31.5) | 96.5              | 88.0    |

Table 5: Split sizes, orthographic token lengths, and lexical diversity. Split rows count collection entries; the final row describes the 610 unique pairs pooled across the benchmark. One pair has membership in three collections. SD denotes sample standard deviation. HD-D uses 42-token samples.

the expected type–token ratio under fixed sampling, reducing the direct dependence on collection size compared with an unadjusted type–token ratio. It characterizes the sampled texts under the specified tokenization.

Tokens are orthographic units obtained after Unicode NFC normalization, with punctuation excluded and Arabic diacritics retained. We apply no Arabic clitic segmentation or lemmatization. Appendix D provides the full convention, counting details, and length quartiles. These statistics contextualize the scores without isolating the effects of length or vocabulary on translation quality.

## 4 Experimental Setup

### 4.1 Evaluated Models

We evaluate twelve systems in three families. The **closed-weight LLMs** are GPT-5 (OpenAI, 2025), Gemini 2.5 Pro (Gemini Team, Google, 2025), and Claude Sonnet 4 (Anthropic, 2025). The **open-weight LLMs** are ALLaM-2-7B (Bari et al., 2024), SILMA-9B-Instruct-v1.0 (SILMA AI, 2024), Qwen2.5-7B-Instruct (Qwen Team,

2025), and Falcon3-10B-Instruct (Falcon-LLM Team, 2024); several target Arabic usage. The **dedicated MT systems** are NAMAA-MT-Saudi2English (NAMAA-MT),<sup>3</sup> NLLB-200-3.3B (NLLB Team, 2022), MADLAD-400-3B (Kudugunta et al., 2023), OPUS-MT for Arabic–English (Tiedemann and Thottingal, 2020), and M2M-100-418M (Fan et al., 2021).

### 4.2 Translation Protocol

We evaluate Saudi→English translation without in-context examples or adaptation on SAFIR. General-purpose LLMs receive a fixed instruction to return only an English translation of the supplied Saudi sentence. The verbatim prompt is in Appendix A. Open-weight LLMs and dedicated MT systems run locally using Hugging Face transformers; the sequence-to-sequence models use a maximum generation length of 256 tokens. This limit may affect the longest entries and is part of the evaluated setup.

<sup>3</sup><https://huggingface.co/Namaa-space/NAMAA-MT-Saudi2English>

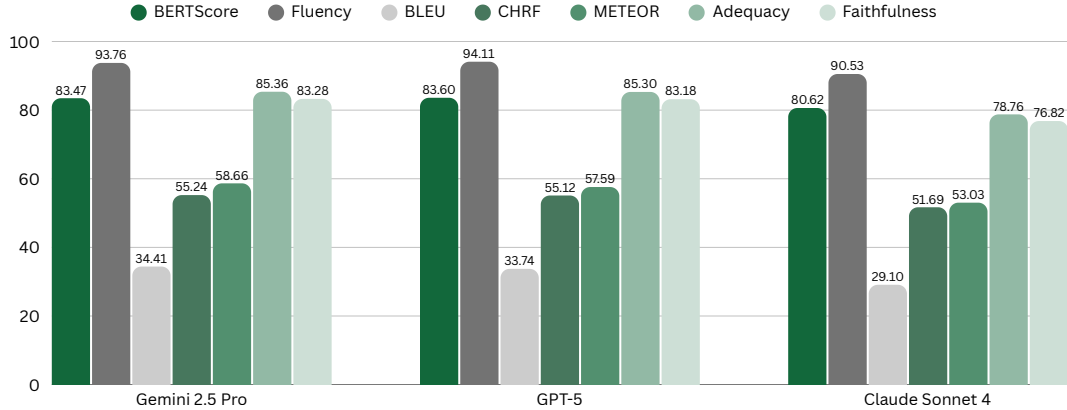


Figure 2: Reported aggregate performance of the three closed-weight LLMs across SAFIR’s 12 evaluation splits. Metrics are on a 0–100 scale; overall BLEU is the mean of the split-level scores.

### 4.3 Automatic Metrics

We use Hugging Face `evaluate` to compute BLEU (Papineni et al., 2002), chrF (Popović, 2015), METEOR (Banerjee and Lavie, 2005), and BERTScore (Zhang et al., 2020). chrF uses the default character  $n$ -gram order, without the word  $n$ -grams of chrF++. BERTScore  $F_1$  uses `deberta-base-mnli` (He et al., 2021), `lang=en`, and no baseline rescaling. All scores are reported on a 0–100 scale.

Overall BLEU is the unweighted arithmetic mean of the twelve split-level BLEU scores, so each split contributes equally. It is not corpus BLEU computed over pooled predictions. Our use of `evaluate` BLEU rather than `sacreBLEU` (Post, 2018) also means that comparisons with other papers require matching implementations and aggregation conventions.

### 4.4 LLM-as-a-Judge

To complement reference-based metrics, we query GPT-4o (OpenAI, 2024) through the OpenAI Batch API for each source, hypothesis, and reference triple. The prompt requests JSON scores from 0 to 100 for adequacy (meaning preservation), faithfulness (omissions and additions), fluency (English naturalness), and holistic overall quality. We report the first three dimensions; Appendix B reproduces the prompt.

These judgments are evaluator-dependent estimates. We neither calibrate them against human ratings of model outputs nor test alternative judges. The human review in Section 3.4 evaluates the benchmark references and does not validate the model-output judge. Close differences between systems should therefore be interpreted cautiously.

## 5 Results and Analysis

We present aggregate results by model family and examine variation across evaluation splits. Appendix E consolidates all twelve systems in Table 14; selected split comparisons appear below.

### 5.1 Closed-Weight LLMs

Figure 2 summarizes the three closed-weight LLMs. Gemini 2.5 Pro obtains the highest average BLEU (34.41), chrF (55.24), and METEOR (58.66), while GPT-5 obtains the highest BERTScore (83.60). Their judged adequacy scores are close: 85.36 and 85.30, respectively. Claude Sonnet 4 obtains 29.10 BLEU and 78.76 adequacy. These point estimates do not establish statistically significant differences.

Rankings change across collections (Table 6). GPT-5 leads BLEU on Eastern Urban (58.20), while Gemini leads on Najdi Qasim (39.59) and Hijazi Jeddah (28.52). Claude leads the three models on Southern Qahtan, scoring 47.68 BLEU against GPT-5’s 42.98 and Gemini’s 43.28. An aggregate ranking therefore does not identify the strongest model on every split.

Reference overlap and judged adequacy can also diverge. Southern Asiri has the lowest adequacy for each closed-weight model (57.10 to 63.20), despite BLEU scores of 27.60 to 34.98. Conversely, Hijazi Urban Jeddah receives adequacy scores of 91.50 to 92.36 with BLEU of 21.76 to 26.58. These cases show that the evaluation dimensions characterize different aspects of output quality; they do not isolate the linguistic cause of either pattern.

The adequacy contrast between Eastern Urban and Southern Asiri is large for each of the three models, ranging from 29.20 to 32.10 points. However,

| Evaluation split    | Claude Sonnet 4 |       | GPT-5 |       | Gemini 2.5 Pro |       |
|---------------------|-----------------|-------|-------|-------|----------------|-------|
|                     | BLEU            | Adeq. | BLEU  | Adeq. | BLEU           | Adeq. |
| Eastern Urban       | 48.33           | 86.30 | 58.20 | 93.00 | 51.06          | 93.70 |
| Hijazi Makkah       | 41.18           | 90.51 | 47.30 | 92.65 | 42.33          | 91.73 |
| Hijazi Urban        | 34.98           | 90.76 | 37.43 | 92.90 | 31.36          | 92.30 |
| Southern Qahtan     | 47.68           | 81.60 | 42.98 | 88.20 | 43.28          | 87.90 |
| Southern Jazan      | 31.04           | 61.70 | 39.82 | 72.50 | 41.67          | 75.70 |
| Najdi Qasim         | 25.22           | 83.50 | 26.95 | 86.60 | 39.59          | 88.50 |
| Najdi Riyadh        | 21.60           | 72.00 | 31.77 | 82.80 | 29.16          | 81.80 |
| Najdi Urban         | 18.46           | 67.40 | 24.02 | 85.61 | 29.28          | 85.10 |
| Hijazi Urban Jeddah | 21.76           | 91.50 | 26.58 | 91.67 | 26.34          | 92.36 |
| Northern Hail       | 15.23           | 82.10 | 16.51 | 85.51 | 19.41          | 85.10 |
| Southern Asiri      | 27.60           | 57.10 | 34.98 | 63.20 | 30.96          | 61.60 |
| Hijazi Jeddah       | 16.17           | 80.70 | 18.28 | 89.00 | 28.52          | 88.50 |

Table 6: BLEU and GPT-4o-judged adequacy (Adeq.) for the three closed-weight LLMs across SAFIR’s 12 evaluation splits. All scores are reported on a scale from 0 to 100.

even this consistent contrast concerns two particular collections and a single judge. A different contrast appears on Hijazi Jeddah: BLEU spans 16.17 to 28.52, while adequacy remains between 80.70 and 89.00. Describing a split as universally “easy” or “hard” would therefore obscure both the evaluation dimension and differences between systems.

## 5.2 Open-Weight LLMs

ALLaM-2-7B leads this family on average BLEU (22.77), adequacy (69.85), and faithfulness (66.72), followed by SILMA and Qwen (Table 7). Falcon3-10B trails on these measures, obtaining 9.97 BLEU and 37.88 adequacy. SILMA has the highest judged fluency in the family (93.06), illustrating that natural English does not necessarily preserve the source meaning.

The leading model also varies by split (Table 8). ALLaM leads the selected comparisons on Eastern Urban, Najdi Urban, and Hijazi Jeddah. SILMA leads on Hijazi Makkah and Southern Qahtan, while Qwen leads on Southern Asiri. These observations describe the evaluated collections; they do not establish a general advantage from Arabic specialization or identify which training data explain the differences.

The selected comparisons also distinguish aggregate strength from local advantages. ALLaM’s Eastern Urban BLEU is 40.21, compared with SILMA’s 34.01 and Qwen’s 31.61. On Southern Asiri, the order reverses: Qwen scores 23.16, SILMA 22.47, and ALLaM 17.17. Falcon is lowest on each listed split. Thus, ALLaM’s aggregate lead coexists with local advantages for other open-weight systems.

## 5.3 Dedicated MT Models

NAMAA-MT has the highest scores among the five dedicated systems on all reported metrics, including 18.53 BLEU and 65.05 adequacy (Table 9). It outperforms NLLB-200, MADLAD-400, OPUS-MT, and M2M-100 under this setup. Although NAMAA-MT targets Saudi Arabic, the comparison does not isolate specialization from other system differences.

Its aggregate advantage does not hold on every collection. NAMAA-MT leads the selected dedicated systems on Eastern Urban and Hijazi Makkah, with BLEU scores of 28.36 and 32.41. On Hijazi Jeddah, OPUS-MT obtains 14.65 BLEU compared with NAMAA-MT’s 8.13 and NLLB’s 3.89. This is another instance in which the overall ordering masks a different local ordering.

M2M-100 obtains 0.00 BLEU on Najdi Urban alongside a fluency score of 67.70 (Table 10). Zero BLEU does not imply zero word overlap: without smoothing, missing higher-order  $n$ -gram matches can produce a zero score. Similarly, OPUS-MT scores 1.84 BLEU and 62.70 fluency on Hijazi Makkah. Fluency concerns English naturalness and cannot establish whether the source meaning is preserved.

## 5.4 Cross-Family Findings

The closed-weight LLMs obtain the strongest aggregate BLEU, adequacy, and faithfulness scores. The remaining families do not form a strict hierarchy: NAMAA-MT exceeds Falcon3-10B on both BLEU and adequacy. The comparison does not isolate the effects of scale, architecture, pretraining coverage, or specialization.

Performance also depends on the sampled collec-

| Model                  | BLEU  | chrF  | METEOR | BERTScore | Adeq. | Faith. | Flu.  |
|------------------------|-------|-------|--------|-----------|-------|--------|-------|
| ALLaM-2-7B             | 22.77 | 44.87 | 48.28  | 78.17     | 69.85 | 66.72  | 82.76 |
| SILMA-9B-Instruct-v1.0 | 21.80 | 42.81 | 44.72  | 77.00     | 66.90 | 63.62  | 93.06 |
| Qwen2.5-7B-Instruct    | 19.75 | 42.01 | 43.05  | 75.96     | 65.55 | 62.42  | 89.54 |
| Falcon3-10B-Instruct   | 9.97  | 30.22 | 30.03  | 68.90     | 37.88 | 33.28  | 83.10 |

Table 7: Reported aggregate performance of the open-weight LLMs. Adeq., Faith., and Flu. denote GPT-4o-judged adequacy, faithfulness, and fluency.

| Evaluation split | ALLaM BLEU | SILMA BLEU | Qwen BLEU | Falcon BLEU |
|------------------|------------|------------|-----------|-------------|
| Eastern Urban    | 40.21      | 34.01      | 31.61     | 11.19       |
| Hijazi Makkah    | 33.75      | 34.47      | 31.15     | 18.19       |
| Najdi Urban      | 20.50      | 18.54      | 12.64     | 6.87        |
| Southern Qahtan  | 33.51      | 34.86      | 32.79     | 17.34       |
| Hijazi Jeddah    | 11.51      | 8.15       | 8.10      | 4.10        |
| Southern Asiri   | 17.17      | 22.47      | 23.16     | 13.93       |

Table 8: BLEU on selected evaluation splits for the four open-weight LLMs. These collections illustrate changes in model ranking; all twelve splits contribute to the aggregate results.

tion and evaluation dimension. Southern varieties show pronounced difficulties in judged adequacy, including the Southern Asiri pattern above. At the same time, creator choices, text length, and reference wording differ across splits. The results support multidimensional reporting for the tested systems and collections, while leaving explanations based on specific linguistic features or pretraining representation to further study.

## 6 Error Analysis

We examine four illustrative outputs to distinguish errors that aggregate metrics can obscure. The examples provide qualitative evidence, rather than estimates of category frequency or comparisons of error rates between model families. They also do not determine the causes of Southern performance gaps.

**Pragmatic intent drift.** For the Hijazi source *إشبك بشويش ليش متصرب*, the output “*What’s wrong? Why are you nervous?*” differs from the reference “*Hey, take it easy. What’s the big hurry?*” The output retains a question about the addressee but loses the calming directive and shifts the interpretation toward an emotional state.

**Speech-act change and invented content.** For Hijazi Jeddah *يا بنت ازهمي أخوكي خليه يجي يتعشا*, the reference is “*Girl, call your brother and have him come eat dinner.*” The output “*Oh, my sister’s son has arrived to visit.*” changes an instruction into a statement about a past event, invents a family relationship, and omits the request to call someone

for dinner.

**Wrong target language and idiom failure.** Najdi Qasim *كفيت ووفيت* has the reference “*You’ve done more than enough.*” The output “*C’est tout, j’ai fini.*” is in French and fails to convey the expression’s appreciative meaning. This combines failure to follow the requested output language with loss of idiomatic intent.

**Lexical substitution and omission.** For Eastern Urban *وين الدريشة أبغى أفتحها يدخل الهوا*, the reference is “*Where’s the window? I want to open it for some air.*” The output “*Where’s the dryer?*” substitutes “dryer” for *الدريشة* (window) and omits the second clause. The result is fluent but both inaccurate and incomplete.

These examples motivate separating lexical errors, omissions, and pragmatic or speech-act changes. A systematic annotation of a larger, model-attributed sample is needed to measure their distribution across varieties.

## 7 Conclusion

SAFIR contributes 610 unique Saudi–English sentence pairs across five regional groups and twelve creator-specific evaluation splits. Native-speaker authorship and translation, together with double review of 235 pairs, support evaluation of the sampled collections. The twelve-system comparison shows strong aggregate performance from closed-weight LLMs, variation in rankings across splits, and differences between reference overlap and judged quality. Southern varieties are particularly challeng-

| Model         | BLEU  | chrF  | METEOR | BERTScore | Adeq. | Faith. | Flu.  |
|---------------|-------|-------|--------|-----------|-------|--------|-------|
| NAMAA-MT      | 18.53 | 41.46 | 46.60  | 76.58     | 65.05 | 61.36  | 78.45 |
| NLLB-200-3.3B | 14.25 | 35.45 | 36.69  | 72.33     | 48.96 | 45.20  | 77.92 |
| MADLAD-400-3B | 11.33 | 31.25 | 33.94  | 69.60     | 45.03 | 41.46  | 76.93 |
| OPUS-MT       | 10.99 | 31.34 | 32.19  | 68.93     | 45.10 | 41.86  | 69.41 |
| M2M-100-418M  | 6.82  | 25.82 | 26.85  | 65.04     | 34.62 | 32.69  | 57.87 |

Table 9: Reported average performance of dedicated MT systems across the 12 evaluation splits. All scores are on a scale from 0 to 100. Adeq., Faith., and Flu. denote GPT-4o-judged adequacy, faithfulness, and fluency, respectively. OPUS-MT denotes Helsinki-NLP/opus-mt-tc-big-ar-en.

| Evaluation split | NAMAA-MT | OPUS-MT | NLLB  | M2M BLEU | M2M Flu. |
|------------------|----------|---------|-------|----------|----------|
| Eastern Urban    | 28.36    | 9.21    | 19.47 | 8.09     | 72.90    |
| Hijazi Makkah    | 32.41    | 1.84    | 27.14 | 17.30    | 65.10    |
| Najdi Riyadh     | 18.44    | 7.91    | 8.65  | 3.59     | 47.80    |
| Najdi Urban      | 10.98    | 6.00    | 8.79  | 0.00     | 67.70    |
| Hijazi Jeddah    | 8.13     | 14.65   | 3.89  | 2.61     | 39.90    |

Table 10: BLEU on selected splits for four dedicated MT systems. The final column reports M2M-100 judged fluency. NLLB and M2M denote NLLB-200-3.3B and M2M-100-418M, respectively.

ing under judged adequacy. These findings motivate broader Saudi coverage and evaluation by variety and quality dimension, without attributing the observed differences to unmeasured linguistic or training-data factors.

## Limitations

SAFIR is a focused evaluation benchmark with limited coverage and uneven representation across regional groups. Its 610 unique pairs do not support population-level generalization to all Saudi speakers or localities. Each split represents one creator, so creator style, text length, and reference wording are confounded with local variety. Some references were prepared with permitted MT assistance, and only 235 pairs received independent double review; low-rated items were retained. The study reports point estimates without confidence intervals or significance tests, so close rankings should not be treated as reliable performance differences. GPT-4o judgments are not validated against human ratings of model outputs, and their robustness to judge choice is untested. Long entries may be affected by generation limits. Findings concern the tested systems and the zero-shot Saudi→English setting. Broader sampling, additional human review, systematic error annotation, judge validation, reverse-direction translation, few-shot prompting, and adaptation remain future work.

## Ethics Statement

Contributors provided informed consent for participation and research release of their contributions. They authored sentences from their knowledge of local language use, rather than collecting material through automated scraping. The benchmark contains no contributor names, usernames, account identifiers, or other personal metadata, and no personal or sensitive information was included in the sentence pairs. Reviewers did not assess their own material. SAFIR is intended for research on translation of sampled Saudi varieties; its split labels and model scores should not be used to rank communities or make claims about the linguistic quality of their speech.

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## A Translation Prompt

All general-purpose LLMs receive the same fixed system message and a per-item user message, in which {text} is replaced by the Saudi source sentence.

### System message

You are a professional translator specializing in Saudi Arabic dialects to English. Provide only the translation, nothing else.

### User message

Translate this Saudi Arabic dialect text to English. Only provide the English translation, no explanations.

Saudi Arabic: {text}  
English:

The reported generation configuration specifies temperature= 0.1 and top\_p= 1 for sampling-based LLM generation. Sequence-to-sequence MT models instead use their model-specific translation interface with a maximum generation length of 256 tokens. These are the shared settings recorded for the experiments; complete per-model API versions and decoding configurations are not reported here.

### B LLM-as-a-Judge Prompt

The GPT-4o judge (Section 4.4) is queried with a fixed *system* message and a per-item *user* message; the braces in the user message are placeholders filled with each (*source*, *hypothesis*, *reference*) triple.

#### System message

You are a bilingual translation evaluator returning only JSON scores.

#### User message

SOURCE (Arabic):  
{source}

HYPOTHESIS (Model Translation):  
{prediction}

REFERENCE (Gold English):  
{reference}

Evaluate the translation on:

- Adequacy (meaning preserved)
- Faithfulness (no omissions / additions)
- Fluency (English grammar / style)
- Overall (holistic correctness)

Return JSON:  
{ "adequacy": 0-100, "faithfulness": 0-100, "fluency": 0-100, "overall": 0-100 }

### C Inter-Annotator Agreement: Coefficient Definitions

**Krippendorff's  $\alpha$  (ordinal).** Krippendorff's  $\alpha$  (Krippendorff, 2004; Artstein and Poesio, 2008) accommodates ordinal scales, two or more raters, and missing values. It is defined as

$$\alpha = 1 - \frac{D_o}{D_e}, \quad (1)$$

where  $D_o$  is the observed disagreement and  $D_e$  the disagreement expected by chance:

$$D_o = \frac{1}{n} \sum_c \sum_k o_{ck} \delta_{ck}^2, \quad (2)$$

$$D_e = \frac{1}{n(n-1)} \sum_c \sum_k n_c n_k \delta_{ck}^2.$$

Here  $o_{ck}$  is the coincidence count for categories  $c$  and  $k$ ,  $n_c = \sum_k o_{ck}$  is the marginal count, and  $n = \sum_c n_c$  is the total number of pairable ratings. For the ordinal metric, the difference function is

$$\delta_{ck}^2 = \left( \sum_{g=\min(c,k)}^{\max(c,k)} n_g - \frac{n_c + n_k}{2} \right)^2, \quad (3)$$

which penalizes distant disagreements more heavily than adjacent ones.

**Gwet's  $AC_2$  (ordinal weights).** Gwet's  $AC_2$  (Gwet, 2014) estimates chance agreement from the overall category proportions in a way that remains stable across a wide range of rating distributions. It is defined as

$$AC_2 = \frac{p_a - p_e}{1 - p_e}, \quad (4)$$

with weighted observed and chance agreement

$$p_a = \frac{1}{n} \sum_{i=1}^n w_{r_{1i}, r_{2i}}, \quad (5)$$

$$p_e = \frac{T_w}{q(q-1)} \sum_{k=1}^q \pi_k (1 - \pi_k),$$

Here  $n$  denotes the number of double-rated items,  $r_{1i}, r_{2i}$  are the ratings of item  $i$ , and  $q = 5$  is the number of scale points,  $\pi_k$  is the proportion of all ratings assigned to category  $k$ , and  $T_w = \sum_{k=1}^q \sum_{l=1}^q w_{kl}$ . We use ordinal (quadratic) weights

$$w_{kl} = 1 - \frac{(k-l)^2}{(q-1)^2}, \quad (6)$$

so that partial credit is awarded to near-misses.

**Descriptive agreement.** For completeness, Table 11 reports the two model-free agreement statistics referenced in Section 3.4.1. For each regional group and each of the three dimensions, authenticity (Auth.), adequacy (Adeq.), and fluency (Flu.), it gives the percentage of the 235 double-reviewed items on which the two reviewers assign exactly the same score (Exact) and the percentage on which

| Group    | N  | Auth. |         | Adeq. |         | Flu.  |         |
|----------|----|-------|---------|-------|---------|-------|---------|
|          |    | Exact | $\pm 1$ | Exact | $\pm 1$ | Exact | $\pm 1$ |
| Najdi    | 59 | 88.1  | 98.3    | 74.6  | 94.9    | 91.5  | 100.0   |
| Northern | 20 | 95.0  | 100.0   | 40.0  | 95.0    | 60.0  | 90.0    |
| Eastern  | 20 | 80.0  | 95.0    | 65.0  | 90.0    | 65.0  | 100.0   |
| Southern | 57 | 89.5  | 96.5    | 75.4  | 91.2    | 80.7  | 94.7    |
| Hijazi   | 79 | 59.5  | 73.4    | 40.5  | 65.8    | 54.4  | 69.6    |

Table 11: Exact and within-one-point ( $\pm 1$ ) agreement as percentages of double-reviewed items within each group.

they agree within one scale point ( $\pm 1$ ). These descriptive figures are interpreted alongside the chance-corrected coefficients of Table 4 in Section 3.4.1.

## D Additional Dataset Statistics

The main text reports geographic assignments and split-level summary statistics in Tables 2 and 5. This appendix specifies the counting and tokenization conventions and provides full length distributions and regional summaries.

### D.1 Counting and Tokenization

The benchmark contains 610 unique Arabic–English pairs. Table 5 reports entries in each creator collection, preserving the split membership used for evaluation. One identical pair occurs in three collections, so split counts are not additive counts of unique pairs. Pooled statistics in the “Unique pairs” row count each distinct pair once. No split was reassigned on the basis of this descriptive audit.

For length and lexical diversity, we apply Unicode NFC normalization and case-fold Latin text. Tokens are contiguous sequences of letters and numbers, retaining combining marks and internal apostrophes while excluding other punctuation. Curly apostrophes are normalized to straight apostrophes. Arabic diacritics are retained; we apply no clitic segmentation, stemming, or lemmatization. Standard deviations use the sample denominator.

Lexical diversity is measured with HD-D at a fixed sample size of 42 tokens (McCarthy and Jarvis, 2010). It is the expected type–token ratio in a sample drawn without replacement, multiplied by 100. For a collection of  $N$  tokens with type frequencies  $f_v$ , we compute

$$\text{HD-D}_{42} = \frac{100}{42} \sum_v \left[ 1 - \frac{\binom{N-f_v}{42}}{\binom{N}{42}} \right]. \quad (7)$$

This statistic characterizes the observed collection

under the stated tokenization; it is not an intrinsic measure of a dialect’s richness.

### D.2 Length Distributions and Regional Summaries

Table 12 reports medians, interquartile ranges, and full ranges for both languages. Table 13 summarizes collection entries by regional group. The large within- and between-collection length differences provide context for interpreting split-level scores, but we do not estimate their independent effect on translation performance.

## E Consolidated Results

Table 14 places all twelve evaluated systems in a common format. Bold values mark the highest reported score for each metric, without implying statistical significance. The main text provides the closed-weight split breakdown and selected comparisons for open-weight LLMs and dedicated MT systems. Complete split-level results are available on the leaderboard.

| Split               | Arabic tokens |            |         | English tokens |           |         |
|---------------------|---------------|------------|---------|----------------|-----------|---------|
|                     | Median        | Q1–Q3      | Min–max | Median         | Q1–Q3     | Min–max |
| Eastern Urban       | 5.5           | 5–6        | 2–9     | 9              | 7.25–10   | 2–16    |
| Hijazi Makkah       | 13            | 7–15       | 4–37    | 19             | 12–23     | 6–58    |
| Hijazi Urban        | 47.5          | 28.25–75.5 | 3–177   | 73             | 39–122.5  | 9–232   |
| Hijazi Urban Jeddah | 21            | 11.5–47    | 5–77    | 36             | 16.5–67.5 | 6–113   |
| Hijazi Jeddah       | 10            | 8–15       | 3–24    | 21             | 16–31     | 5–44    |
| Najdi Qasim         | 11            | 7–16       | 2–106   | 16             | 11.5–24.5 | 5–175   |
| Najdi Riyadh        | 8             | 5–12       | 1–39    | 13             | 7.5–20    | 1–63    |
| Najdi Urban         | 4             | 3–5        | 2–11    | 6              | 5–8       | 1–16    |
| Northern Hail       | 6             | 5–7        | 3–9     | 8              | 6.25–10   | 4–14    |
| Southern Qahtan     | 6             | 5–8        | 3–26    | 11             | 7–13      | 4–47    |
| Southern Jazan      | 4             | 3–4.75     | 2–6     | 6.5            | 4.25–8    | 3–13    |
| Southern Asiri      | 6.5           | 1–15.75    | 1–28    | 11             | 2–20.5    | 1–40    |

Table 12: Length distributions in each creator collection. Q1 and Q3 are the first and third quartiles. Length is measured using the orthographic tokenization specified in Appendix D.

| Group    | Entries | Mean length (SD) |             | Median length |         | HD-D $\times 100$ |         |
|----------|---------|------------------|-------------|---------------|---------|-------------------|---------|
|          |         | Arabic           | English     | Arabic        | English | Arabic            | English |
| Eastern  | 54      | 5.6 (1.4)        | 8.9 (2.6)   | 5.5           | 9       | 94.2              | 84.1    |
| Hijazi   | 201     | 27.3 (28.4)      | 42.3 (41.3) | 16            | 28      | 96.2              | 87.7    |
| Najdi    | 154     | 11.4 (17.4)      | 18.4 (28.9) | 7             | 10      | 96.1              | 88.4    |
| Northern | 50      | 5.9 (1.4)        | 8.3 (2.4)   | 6             | 8       | 93.4              | 85.0    |
| Southern | 153     | 6.7 (6.0)        | 10.3 (8.5)  | 5             | 8       | 94.5              | 86.6    |

Table 13: Regional summaries of collection entries. Entries preserve split membership; they are not deduplicated counts. Lexical diversity is computed after pooling each group’s entries, rather than averaging its split scores.

| Model                      | Reference-based metrics |              |              |              | GPT-4o judgments |              |              |
|----------------------------|-------------------------|--------------|--------------|--------------|------------------|--------------|--------------|
|                            | BLEU                    | chrF         | METEOR       | BERTScore    | Adeq.            | Faith.       | Flu.         |
| <i>Closed-weight LLMs</i>  |                         |              |              |              |                  |              |              |
| Gemini 2.5 Pro             | <b>34.41</b>            | <b>55.24</b> | <b>58.66</b> | 83.47        | <b>85.36</b>     | <b>83.28</b> | 93.76        |
| GPT-5                      | 33.74                   | 55.12        | 57.59        | <b>83.60</b> | 85.30            | 83.18        | <b>94.11</b> |
| Claude Sonnet 4            | 29.10                   | 51.69        | 53.03        | 80.62        | 78.76            | 76.82        | 90.53        |
| <i>Open-weight LLMs</i>    |                         |              |              |              |                  |              |              |
| ALLaM-2-7B                 | 22.77                   | 44.87        | 48.28        | 78.17        | 69.85            | 66.72        | 82.76        |
| SILMA-9B-Instruct-v1.0     | 21.80                   | 42.81        | 44.72        | 77.00        | 66.90            | 63.62        | 93.06        |
| Qwen2.5-7B-Instruct        | 19.75                   | 42.01        | 43.05        | 75.96        | 65.55            | 62.42        | 89.54        |
| Falcon3-10B-Instruct       | 9.97                    | 30.22        | 30.03        | 68.90        | 37.88            | 33.28        | 83.10        |
| <i>Dedicated MT models</i> |                         |              |              |              |                  |              |              |
| NAMAA-MT                   | 18.53                   | 41.46        | 46.60        | 76.58        | 65.05            | 61.36        | 78.45        |
| NLLB-200-3.3B              | 14.25                   | 35.45        | 36.69        | 72.33        | 48.96            | 45.20        | 77.92        |
| MADLAD-400-3B              | 11.33                   | 31.25        | 33.94        | 69.60        | 45.03            | 41.46        | 76.93        |
| OPUS-MT (Arabic–English)   | 10.99                   | 31.34        | 32.19        | 68.93        | 45.10            | 41.86        | 69.41        |
| M2M-100-418M               | 6.82                    | 25.82        | 26.85        | 65.04        | 34.62            | 32.69        | 57.87        |

Table 14: Reported average performance across the 12 evaluation splits. All scores are on a scale from 0 to 100. Adeq., Faith., and Flu. denote adequacy, faithfulness, and fluency, respectively. Bold indicates the highest reported value in each column, without implying statistical significance. NAMAA-MT denotes NAMAA-MT-Saudi2English.